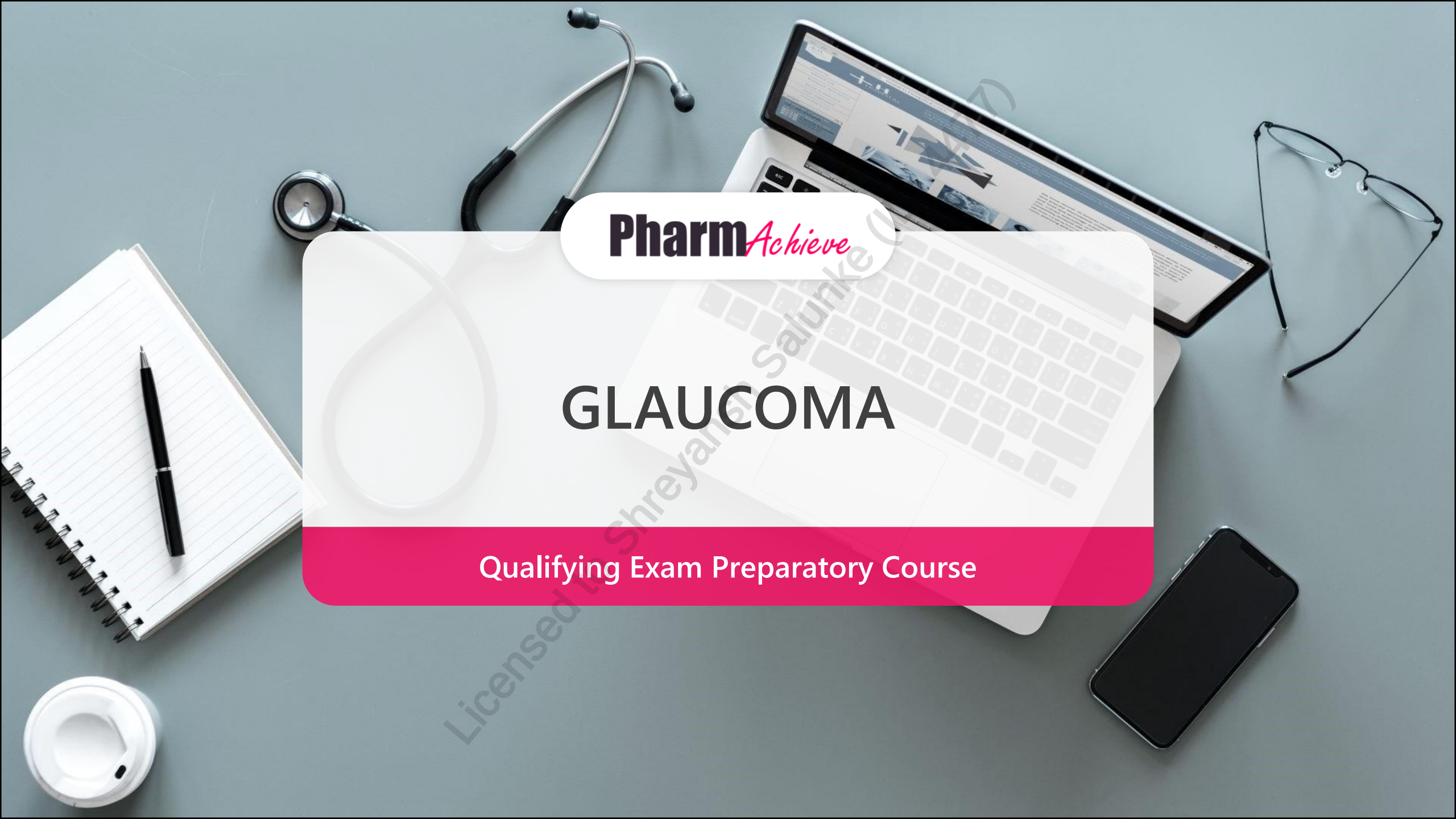




Pharm*Achieve*

GLAUCOMA

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

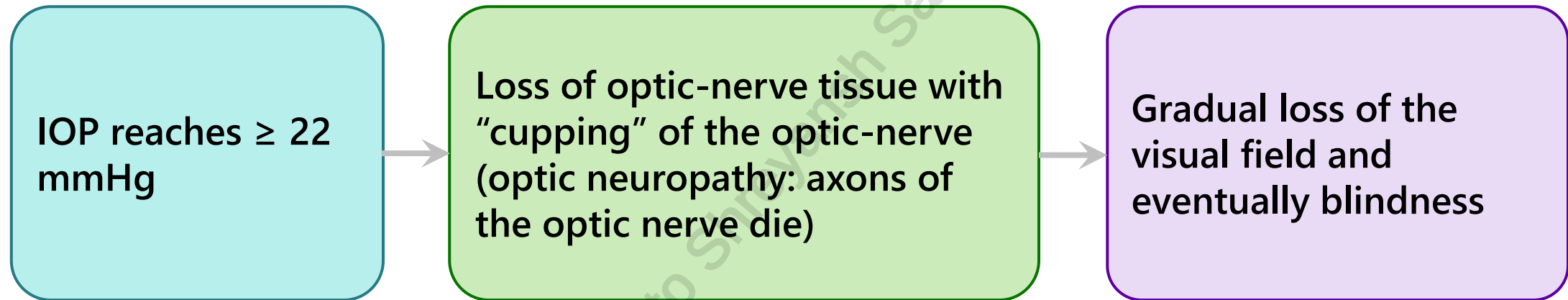
AE	Adverse Effect
BB	Beta Blocker
BMI	Body Mass Index
CNS	Central Nervous System
GI	Gastrointestinal
H	Hours
HS	Hora Somni (At Bedtime)
IOP	Intraocular pressure
NLO	Nasolacrimal Occlusion
PO	Per Os (Oral)
Q	Quaque (Every)
QAM	Quaque Die Ante Meridiem (Every Morning)
UV	Ultraviolet

LEARNING OBJECTIVES

- 1 Define glaucoma and explain the pathophysiology
- 2 Outline risk factors and drug-induced causes of glaucoma
- 3 Identify drugs used for open-angle glaucoma
- 4 Elaborate on mechanisms of action and side effects of medications

WHAT IS GLAUCOMA?

- Progressive, irreversible optic nerve damage that results in visual field loss and eventually blindness if left untreated
- Nerve damage is caused by elevated intraocular pressure (IOP) within the eye



PATHOPHYSIOLOGY

Ciliary body makes aqueous humour (clear fluid)

Aqueous humour goes into anterior chamber of eye (a space between iris and cornea)

Aqueous humour then travels through drainage angle, then trabecular meshwork and then through Schlemm's canal

When the rate of aqueous humour production is not balanced with rate of outflow → ↑ IOP → optic nerve damage

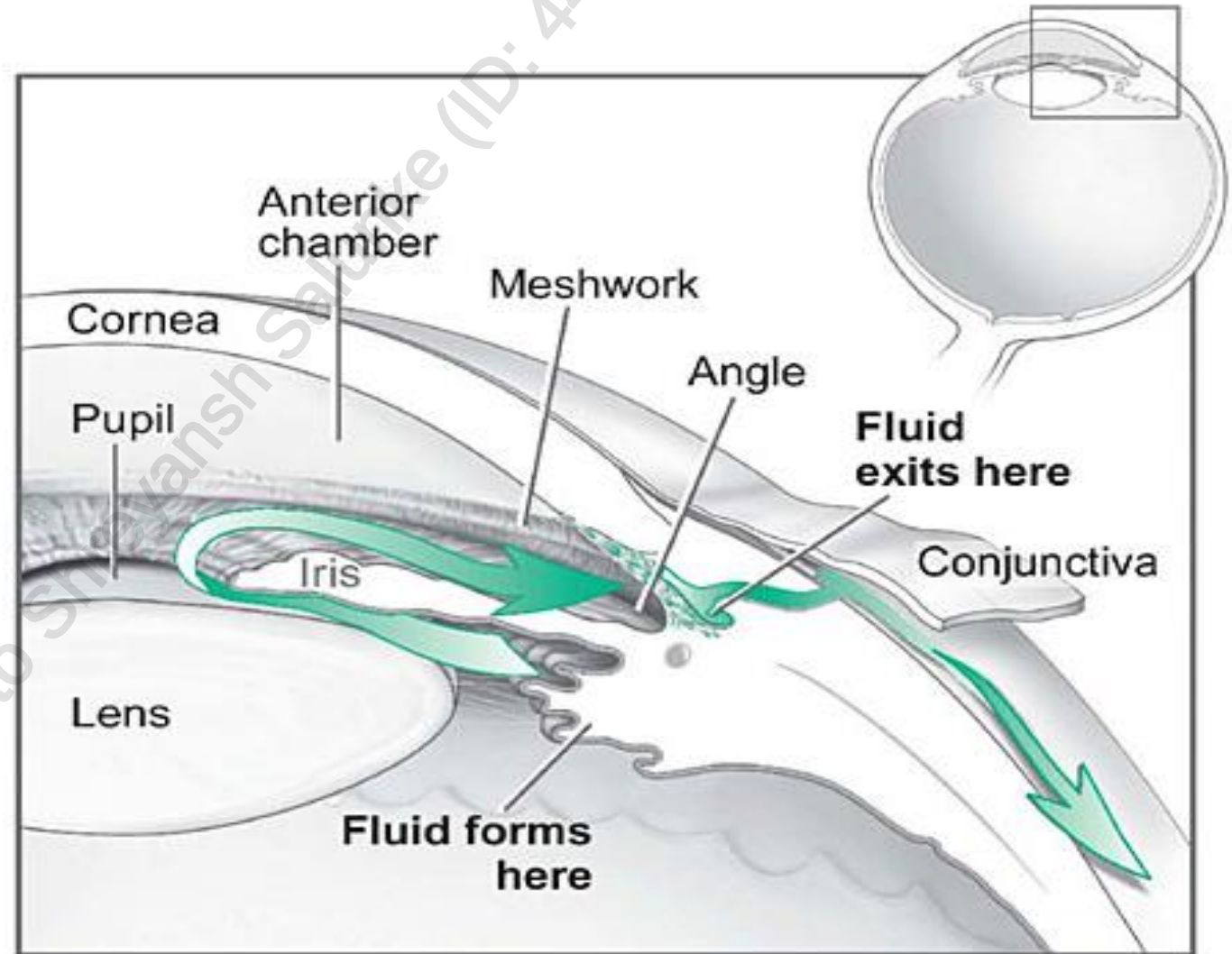
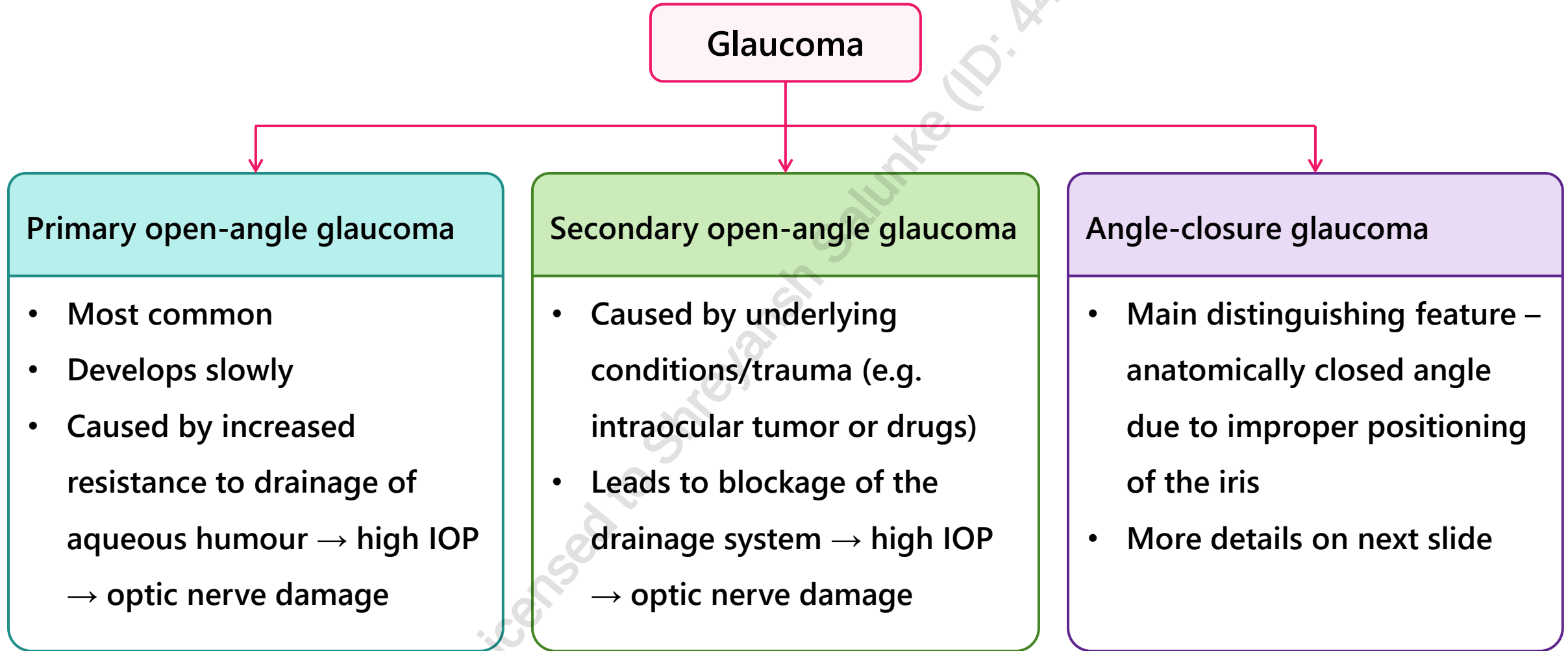


Image Courtesy of the NIH, National Eye Institute

TYPES OF GLAUCOMA



ANGLE-CLOSURE GLAUCOMA

✓ Can be chronic or acute (< 1/3 of cases)

Chronic Angle-Closure Glaucoma

- Intermittent obstruction of angle
- Symptomatic or asymptomatic
- Over time, can cause potential damage to optic nerve, which prevents fluid drainage
- Mainstay of treatment: laser or surgical iridectomy + aggressive medical treatment to ameliorate damaging effects of IOP elevation until surgery is performed

Acute Angle-Closure Glaucoma

Medical
Emergency!



- Pupillary block of the iris → full angle closure → obstruction of aqueous humour outflow → sudden rapid increase in IOP
- Symptoms: sudden blurred vision, severe pain, halos formed around bright lights, nausea and vomiting
- Diagnostic characteristics: significantly elevated IOP (>30 mmHg , corneal swelling, conjunctival hyperemia (dilatation of conjunctival blood vessels)
- May terminate without treatment, but if IOP remains high, it will cause irreversible optic nerve damage

RISK FACTORS

Primary open-angle glaucoma	Elevated IOP, age >40 years, ethnicity (African descent e.g. African Americans), positive family history, myopia (near-sightedness), vascular disease (e.g., migraines, hypertension), type 2 diabetes mellitus, lifestyle measures (e.g. smoking, obesity)
Secondary open-angle glaucoma	Trauma (blunt or penetrating), prior intraocular inflammation/surgery, corticosteroid use, pseudoexfoliation syndrome (accumulation of microscopic granular amyloid-like protein fibers in the eye), pigment dispersion syndrome (pigment rubs off the back of the iris)
Angle-closure glaucoma	Ethnicity (Asian, Inuit), female sex, advanced age, hyperopia (blurred vision), history of angle-closure in contralateral eye, positive family history, anatomical risk factors (e.g. smaller volume/depth of anterior chamber i.e. shallow chamber)

DIFFERENTIAL DIAGNOSIS

Ocular Hypertension

- IOP > 21 mmHg but normal visual fields and normal optic discs

Diseases of the optic nerve or retina

- Can have overlapping visual defects
- E.g. ischemic optic neuropathy, retinitis pigmentosa, etc.

DRUG-INDUCED CAUSES

*Corticosteroids

Systemic, ophthalmic, nasal, inhaled

If IOP develops,
corticosteroids must
be tapered ASAP

Anticholinergic agents

Antihistamines, decongestants, antidepressants,
antispasmodics

Topiramate

Associated with acute angle-closure glaucoma

Oral contraceptives

Use of longer than 3-5 years may be associated with
higher prevalence of primary open-angle glaucoma

GOALS OF THERAPY

- ✓ Prevent or slow progressive visual loss
- ✓ Preserve the function and structure of the optic nerve
- ✓ Eliminate pain and improve vision loss in acute forms
- ✓ Improve vision and quality of life

NON-DRUG MEASURES

- Lifestyle modifications have not shown to alter the outcome of the disease
- Aerobic exercise can lower IOP in some patients to a modest degree
- Laser or surgical procedures when drug therapy is unsuccessful

PHARMACOLOGICAL THERAPY



Topical agents preferred due to ↓ systemic side effects and ↑ concentration at the site of action

Topical Agents

- **Prostaglandin analogs** – 1st-line
- **Beta-blockers** – 2nd-line

Add/Substitute:

- Carbonic anhydrase inhibitors
- Alpha₂-adrenergic agonists
- Cholinergic agonists

Combination Products (topical)

- Various combinations of two topical agents

Oral Agent

- Carbonic anhydrase inhibitors

PHARMACOLOGICAL THERAPY

PROSTAGLANDIN ANALOGUES

Drugs	Latanoprost, Travoprost, Bimatoprost, Latanoprostene bunod
MOA	Latanoprost and travoprost: ↓ intraocular pressure by ↑ outflow of aqueous humour through uveoscleral pathway. Bimatoprost and latanoprostene bunod: ↑ uveoscleral and trabecular meshwork outflow
Efficacy	1st line due to their excellent safety profile and effectiveness. Slightly more effective than beta-blockers
AEs	Foreign body sensation, burning, stinging, itching, increased iris pigmentation, increased eyelash length, headache, flu-like symptoms, conjunctival hyperemia (redness)
Tips	Latanoprost: store in fridge until dispensed (6 weeks at room temperature)

PHARMACOLOGICAL THERAPY

BETA-BLOCKERS

Drugs	Timolol
MOA	↓ IOP by inhibiting the formation of aqueous humour, visual field protection
C/I	Pulmonary and cardiac disease* * Depending on physician's assessment
AEs	• Local Effects: stinging, dry eyes, conjunctivitis (rare) • Systemic Effects: bronchospasms, exacerbation of asthma, congestive heart failure, dyspnea, bradycardia, hypotension, syncope, depression, impotence, altered response to hypoglycemia, reduction of high-density lipoproteins
Tips	• Slightly less effective than prostaglandin analogues • Use with caution in patients with history of syncope or bradycardia • Timolol = 1 drop QAM or Q12H • Timolol gel-forming solution = 1 drop once daily • Systemic side effects reduced using nasolacrimal duct occlusion technique

PHARMACOLOGICAL THERAPY

TOPICAL CARBONIC ANHYDRASE INHIBITORS

Drugs	Brinzolamide, Dorzolamide
MOA	↓ IOP by inhibiting the enzyme carbonic anhydrase, which is involved in the production of aqueous humour
Efficacy	• Not as effective at lowering IOP as compared to first-line • Place in therapy: Adjunctive therapy or primary treatment for those with intolerances or contraindications to prostaglandin analogues or BBs
AEs	Bitter, sour, or unusual taste, stinging, local allergic reaction [cross-sensitivity with sulfa-allergy]

PHARMACOLOGICAL THERAPY

ALPHA₂-ADRENERGIC AGONISTS

Drugs	Brimonidine, Apraclonidine
MOA	↓ IOP by suppressing the formation of aqueous humour and ↑ uveoscleral outflow
C/Is	Monoamine oxidase inhibitors
AEs	Local allergic reaction (allergic conjunctivitis), ocular itching, dry mouth, tachycardia, hypotension, headache, tremor
Tips	• Due to higher rate of local allergic reactions, apraclonidine is limited to management of acute IOP spikes • Brimonidine has higher alpha₂ selectivity and is used for both open and closed angle glaucoma

PHARMACOLOGICAL THERAPY

CHOLINERGIC AGONISTS

Drugs	Pilocarpine
MOA	Stimulates muscarinic receptors to contract the ciliary muscle and ↑ trabecular outflow
Efficacy	3 rd line due to side effect profile and frequency of administration
AEs	Reduced vision in patients with cataracts, blurred vision due to refractive shift, brow ache, miosis, myopia, GI upset (rare), retinal tear/detachment (rare)
Tips	Poorly tolerated in children and young adults

PHARMACOLOGICAL THERAPY

ORAL CARBONIC ANHYDRASE INHIBITORS

Drugs	Acetazolamide, Methazolamide
MOA	↓ IOP by inhibiting the enzyme carbonic anhydrase, which is involved in the production of aqueous humour
C/Is	Hepatic dysfunction, renal insufficiency
AEs	Paresthesia of the extremities, metabolic acidosis, hypokalemia, GI upset, urolithiasis, lethargy and depression, aplastic anemia (rare), Stevens-Johnson syndrome (rare)
Tips	• Methazolamide has less severe AEs • Reserved for medical emergencies due to significant side effects • 50% of patients cannot tolerate long term due to GI/CNS difficulties, paresthesias, metabolic acidosis or renal lithiasis

PRESERVATIVES

- To prevent growth of microorganisms
- Consider patients' allergies to preservatives and adjust therapy accordingly
- Benzalkonium chloride denatures proteins and causes lysis of cytoplasmic membranes. If used for too long, can damage ocular surface and cause irritation, dryness, itching and burning
- Ionic buffer preservative
- Purite degrades to chloride ions and water when exposed to UV light
- Preservative-free, single-dose units

MONITORING PARAMETERS

EFFICACY ENDPOINTS

Parameter	Degree of Change	Timeframe
IOP	Reduced	Within 2 weeks (Prostaglandin analogues may take up to 4-6 weeks for maximal effect)

SAFETY ENDPOINTS

Parameter	Degree of Change	Timeframe
Common ADRs e.g. ocular burning/stinging	Minimal (tolerable)	Duration of therapy
Serious/uncommon ADRs	None	Duration of therapy

- Ensure patient is using nasolacrimal duct occlusion method when instilling drops to prevent systemic adverse drug reactions
- Refer patient to a physician if any of the following occur: Unexplained eye pain, vision loss, abnormal photophobia, eye irritation or redness from eye drops that becomes bothersome, changes in comorbid conditions (asthma/COPD, hypotension), systemic symptoms

COUNSELLING TIPS

Nasolacrimal Duct Occlusion (NLO)

Blocking the nasolacrimal drainage duct by placing a finger on it during eye drop administration

- Without NLO ~80% of a dose will flow into nasolacrimal drainage apparatus
- ~30-80% of the drug is absorbed into the systemic circulation causing systemic side effects
- NLO can ↓ systemic absorption & ↑ contact time with the cornea
 - ↓ systemic side effects and ↑ ocular bioavailability of the drug → ↑ medication effectiveness
- Important to educate about nasolacrimal/ punctal occlusion during administration of eye drops

COUNSELLING TIPS

- **Suspension** eye drops should be **shaken well** before use
- Remove contact lenses prior to instilling drops & wait at least 15 min after instilling drops to insert contact lens
- If the applicator of the drop touches a surface, wipe with a clean tissue. Do not rinse tip of bottle with water as this will change the solution's stability and composition

1. Wash hands and pull-down lower eyelid using thumb and index finger to **form a pouch**
2. Instill only 1 drop and close eye without squeezing or blinking for 30-60 seconds to allow absorption—use **nasolacrimal duct occlusion** by applying gentle pressure with finger across the bridge of the nose to prevent systemic absorption
3. **Wait 3-5 min between administration of different eyedrops** to prevent flushing away or diluting the first drop—if you are using both eye drops and eye ointment, apply the eyedrop at least 10 minutes before the ointment.
4. Remove excess fluid gently with a clean tissue—do not rub
5. Discard bottle **after 28 days** after opening



Refer to the product monograph for specific instructions on proper use, as information may vary

TAKEAWAY

Glaucoma is the progressive, irreversible damage to the optic nerve causing visual field loss and possibly blindness if left untreated. It is characterized by an elevated intraocular pressure (IOP)

Normal IOP is 10-21 mmHg

Topical agents are preferred to reduce the risk of systemic side effects and increase the concentration at the site of action. Combination products are available

Prostaglandin analogues are **1st line** due to their safety profile and efficacy (slightly more effective than beta-blockers). They reduce IOP by increasing aqueous humour outflow

Timolol is a **2nd line** agent that reduces IOP by inhibiting the formation of aqueous humour. Avoid in cardiac and pulmonary disease (may be used at the discretion of the prescriber)

Alternative/additional agents include **carbonic anhydrase inhibitors** (topical or PO), **alpha₂-adrenergic agonists**, and **cholinergic agonists**

CASE STUDY

GD is a 41-year-old female who was referred to the clinic by her optometrist as she has been experiencing progressive peripheral vision loss in both eyes. GD reports occasional mild headaches and difficulty seeing at night, however, she denies any pain in her eyes. GD has a past medical history that is significant for migraine headaches without aura, allergic rhinitis, hypertension, and type 2 diabetes mellitus. Her current medications include topiramate 25 mg PO daily (initiated 6 months ago), amlodipine 2.5 mg PO daily (initiated 1 year ago), perindopril 8 mg PO daily (initiated 1.5 years ago), metformin 1000 mg PO BID (initiated 5 years ago), empagliflozin 10 mg PO daily (initiated 3 years ago), loratadine 10 mg PO daily (initiated 2 years ago), and ethinyl estradiol 20 mcg/levonorgestrel 0.1 mg PO daily (initiated 6 years ago). GD wears glasses everyday as she has been near-sighted for as long as she can remember. She weighs 194 lbs and is 167 cm tall. She denies any alcohol, smoking, or recreational drug use. GD reports that she drinks 3 cups of caffeine daily. Her intraocular pressure (IOP) is recorded as 26 . GD is diagnosed with primary open-angle glaucoma and prescribed latanoprost 0.005% ophthalmic solution 1 drop instilled into both eyes once daily HS. She has no known allergies to any medications.

CASE STUDY - QUESTIONS

1. Which of the following is NOT a risk factor GD has for primary open-angle glaucoma?
 - a) Myopia
 - b) Body mass index (BMI)
 - c) Female sex
 - d) Migraine history
2. Which of the following drugs is NOT associated with the development of glaucoma?
 - a) Topiramate
 - b) Ethinyl estradiol/levonorgestrel
 - c) Metformin
 - d) Loratadine
3. Which of the following is NOT an appropriate counselling point to review with GD?
 - a) Store this medication in the fridge after opening
 - b) Monitor for adverse effects such as eye lash lengthening, iris pigmentation, and redness
 - c) Aerobic exercise can help to lower the IOP
 - d) Close the eyes gently for 30-60 seconds after administering each drop
4. Which of the following agents is LEAST appropriate to add-on to further lower the IOP for GD?
 - a) Timolol
 - b) Methazolamide
 - c) Brimonidine
 - d) Dorzolamide

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CHANGE LOG

March 2025

- Slide 3: Updated legend
- Slide 8: Changed 'chronic closure angle' to 'chronic angle-closure..'
- Slide 24: Changed 5-10 mins to 3-5 mins and added disclaimer at bottom that instructions may vary- consult product monograph

August 2025

- Formatting, grammar, spelling changes throughout
- Slide 3: Updated legend
- Slide 9: Changed 'gender' to 'sex' under angle-closure glaucoma
- Slide 16: Removed betaxolol
- Slide 25: Added takeaway
- Slides 26, 27: Added case study and questions

November 2025

- No content changes made

February 2026

- Updated NAPRA competencies slide
- Minor formatting changes made throughout
- Slide 22: Updated monitoring parameters
- Slide 29: Updated references